WakeBand flex rev A: 2-layer polyimide flex, folds once around a 12 x 30 mm 110 mAh cell.
Pod about 22 x 36 x 8 mm. Sheets connect through global labels and power nets.
Status: step 1 (project) and step 2 (ISP2454 symbol + GPIO plan) done; other sheets are notes.

power

Power: nPM1304 PMIC, battery, NTC, ship mode, current jumper

File: power.kicad_sch

mcu

MCU: ISP2454 (nRF54L15) module, SWD, UART, I2C, interrupts

File: mcu.kicad_sch

sensors

Sensors: LIS2DW12 accelerometer

File: sensors.kicad_sch

haptics

Haptics: DRV2605L driver and LRA pads

File: haptics.kicad_sch

dock

Dock: six pogo pads, TVS, series resistors

File: dock.kicad_sch

debug

Debug: test pads, debug LED (DNP option)

File: debug.kicad_sch

Plan: docs/flex-pcb-schematic-plan.md

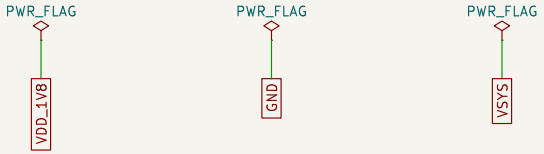
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Sheet: /
File: flex-rev-a.kicad_sch

Title: WakeBand flex rev A — Top

Size: A4	Date: 2026-10-30	**Rev: A0**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/7

Power sheet (step 3, drawn after the Stage 0b nPM1304-EK tests):
- nPM1304 QFN32: VBUS from dock pad VIN (TVS), VBAT to cell pads through the current-measure jumper,
  BUCK1 = 1.8 V (VSET1 resistor), BUCK2 unused (VOUT2 to VSYS, VSET2 to GND), VDDIO to VOUT1.
- 10 k NTC on the cell to the NTC pin, JEITA hot cut-off about 45 C.
- SHPHLD pull-up + test pad + optional button footprint. PMIC_INT to the SoC.
- VSYS feeds the DRV2605L.
TEMPORARY: the two PWR_FLAGs below stand in for the nPM1304 outputs so ERC passes. Remove them in step 3.



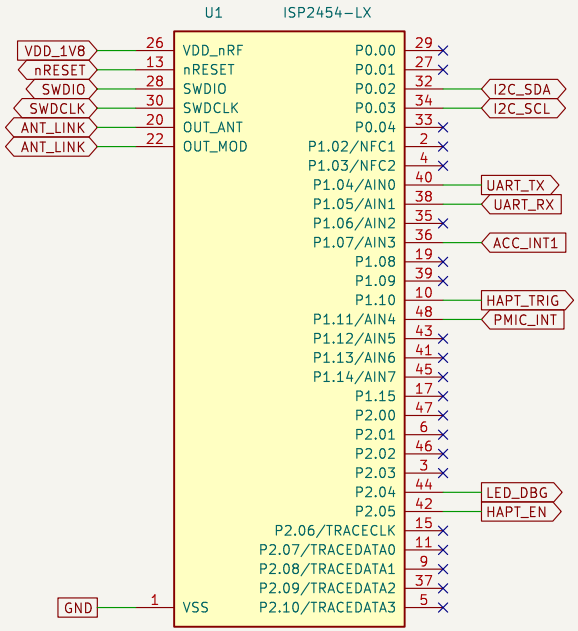
Plan: docs/flex-pcb-schematic-plan.md
FOSS Health Alarm

Sheet: /power/
File: power.kicad_sch

Title: WakeBand flex rev A — Power

Size: A4	Date: 2026-10-30	**Rev: A0**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 2/7

ISP2454 notes (datasheet R7):
- Antenna, 32 MHz + 32.768 kHz crystals, DC-DC inductor and all decoupling are inside. No external parts.
- OUT_ANT (20) must be joined to OUT_MOD (22) by a short trace (net ANT_LINK).
- Pads 49-62 are mechanical: isolated copper in the footprint, not in this symbol.
- Antenna corner 1 mm from the board edge, no copper or parts on any layer in the keep-out, including the folded battery layer below it.
- Firmware: set LFXO load capacitance to 19 pF (devicetree load-capacitance-femtofarad = <19000>).
- Every signal uses an OUTER-row pad, so the flex needs no via-in-pad. Inner GND pads (21, 23) and nRESET (13) may use tented vias under the module centre, as the datasheet allows.
- I2C on TWIM30 (P0, low-power domain). UART on UARTE20 (P1).
- P1.02/P1.03 are NFC pins: left unused. P2 trace pins left unused.
- GPIO plan and reasons: docs/flex-rev-a-pinout.md



Plan: docs/flex-pcb-schematic-plan.md

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Sheet: /mcu/
File: mcu.kicad_sch

Title: WakeBand flex rev A — MCU

Size: A4	Date: 2026-10-30	**Rev: A0**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 3/7

Sensors sheet (step 5): LIS2DW12 LGA-12 on VDD_1V8.
- CS tied to VDD_1V8 (I2C mode), SA0 strapped high -> address 0x19.
- SDA/SCL on I2C_SDA / I2C_SCL, 4.7 k pull-ups to VDD_1V8 (shared bus).
- INT1 -> ACC_INT1 (wake), INT2 -> ACC_INT2.
- 100 nF on VDD and VDD_IO.

Plan: docs/flex-pcb-schematic-plan.md
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Sheet: /sensors/
File: sensors.kicad_sch
Title: WakeBand flex rev A — Sensors

Size: A4	Date: 2026-10-30	**Rev: A0**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 4/7

Haptics sheet (step 5): DRV2605L VSSOP-10 (DRV2605LDGS) on VSYS.
- 1 uF on VDD and on REG. EN <- HAPT_EN, IN/TRIG <- HAPT_TRIG.
- I2C at 1.8 V (VIH 1.3 V), address 0x5A. Keep VDD powered (an unpowered driver clamps the bus).
- OUT+/OUT- to two LRA lead pads beside the stiffened 9 mm LRA tape-down pad.
- Firmware blocks the motor while docked (VSYS can reach 5.5 V).

Plan: docs/flex-pcb-schematic-plan.md
FOSS Health Alarm
Sheet: /haptics/
File: haptics.kicad_sch
Title: WakeBand flex rev A — Haptics
Size: A4 | Date: 2026-10-30 | **Rev: A0**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 | Id: 5/7

Dock sheet (step 6): six pads at 2 mm pitch, about 1.2 x 3 mm:
1 VIN 2 GND 3 SWDCLK 4 SWDIO 5 UART_RX 6 UART_TX
- TVS on every pad, leakage under 50 nA; VIN TVS working voltage at least 5.5 V.
- 100 ohm in series on the four data pads. Pull-down on UART_RX.

Plan: docs/flex-pcb-schematic-plan.md
FOSS Health Alarm
Sheet: /dock/
File: dock.kicad_sch
Title: WakeBand flex rev A — Dock
Size: A4 | Date: 2026-10-30 | **Rev: A0**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 | Id: 6/7

Debug sheet (step 6): test pads for every signal plus nRESET, SHPHLD, ACC_INT2 and spare GPIOs; one 0402 LED on LED_DBG with a DNP option.

Plan: docs/flex-pcb-schematic-plan.md

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Sheet: /debug/

File: debug.kicad_sch

Title: WakeBand flex rev A — Debug

Size: A4	Date: 2026-10-30	**Rev: A0**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 7/7